

MAT3032 Unassessed Coursework 2 Answers

1. (a) If $a, b \in H, x \in G$ then $a, b, x \in G$, so $(a, b)(x) \in G$.
 (e, e) is the identity of $H \times H$, and $(e, e)(x) = exe = x$ for all $x \in G$.
Let $(a, b), (c, d) \in H \times H$. Then $(a, b)((c, d)(x)) = (a, b)(cxd) = acxdb$
As H is abelian, $acxdb = acxbd = (ac, bd)(x) = ((a, b)(c, d))(x)$.
Hence $(a, b)(x) = axb$ defines an action of $H \times H$ on G . [5]

- (b) $\mathcal{O}(e) = \{aeb : a, b \in H\}$. Since $e \in H$, for any $a \in H$ we have $a = aee \in \mathcal{O}(e)$, so $H \subset \mathcal{O}(e) \subset H$, so $\mathcal{O}(e) = H$.
 $(H \times H)_e = \{(a, b) \in H \times H : aeb = e\} = \{(a, a^{-1}) : a \in H\}$.
 $\text{Fix}(e, e) = \{x \in G : exe = x\} = G$. [4]

2. (a) $x^2y = x(xy) = (yxy)(xy) = yx(yxy) = (yx)x = yx^2$, so x^2 commutes with y . Also $x^2x = x^3 = xx^2$ so x^2 commutes with x . As it commutes with the generators, x^2 commutes with all elements of G so $x^2 \in Z(G)$. [4]

- (b) By (a), $\{x^2\}$ is a conjugacy class. So is $\{e\}$.
 $xyx^{-1} = y^{-1} = y^2$, so y and y^2 are in the same conjugacy class. As they are the only two elements remaining, $\{y, y^2\}$ is a conjugacy class.
Alternatively, $yx = xy^{-1} \neq xy$, so $y \notin Z(G)$. Thus $\{y\}$ is not a conjugacy class, so y and y^2 must be in the same class. [4]

- (c) A subgroup is normal if it is a union of conjugacy classes. Now $\{e\} \cup \{x^2\} = \{e, x^2\}$ is a subgroup since $(x^2)^2 = x^4 = e$ (or since it is $Z(G)$), and $\{e\} \cup \{y, y^2\} = \{e, y, y^2\} = \langle y \rangle$ which is also a subgroup, so these are normal subgroups of orders 2 and 3. [4]

- (d) $Z(G) = \{e, x^2\}$ so $G/Z(G)$ has order $\frac{12}{2} = 6$. As G is non-abelian, $G/Z(G)$ is not cyclic by Proposition 2.10, so it must be isomorphic to S_3 (or equivalently D_3). [3]

3. σ has order $\text{lcm}(2, 2, 3) = 6$, cycle index 3 and cycle type $2^2 3^1$.
6 is also the order of permutations in S_7 of type $1^2 2^1 3^1$ and $1^1 6^1$.
By Proposition 2.15, the required number is $\frac{7!}{2^2 3^1 2!} + \frac{7!}{1^2 2^1 3^1 2!} + \frac{7!}{1^1 6^1}$
($= 1470$). [6]

4. (a) The only symmetries in the plane are the identity $(1)(2)(3)(4)$ and the rotations $(1\ 2\ 3)(4)$ and $(1\ 3\ 2)(4)$.

The symmetries form a group of order 3, so by Proposition 2.18 the number of q -colourings is $\frac{1}{3}(q^4 + 2q^2)$. [4]

- (b) Now we also have the reflections $(1\ 2)(3)(4)$, $(1\ 3)(2)(4)$, $(2\ 3)(1)(4)$.

The group has order 6, so the number of q -colourings is $\frac{1}{6}(q^4 + 3q^3 + 2q^2)$. [4]

5. (a) $w_2 \equiv_2 1$ and $w_2 \mid 49$, so $w_2 \in \{1, 7, 49\}$.

$w_7 \equiv_7 1$ and $w_7 \mid 8$, so $w_7 \in \{1, 8\}$.

The 2-Sylows have order 8 and the 7-Sylows have order 49. [4]

- (b) G has at least one 7-Sylow, of index 8, so if G is simple then by Proposition 2.29 it is isomorphic to a subgroup of S_8 . But $|G|$ does not divide $8!$, so G cannot be simple. [4]

- (c) If $w_2 = w_7 = 1$ then by Proposition 2.22 (iii) $G \cong P \times Q$ where the 2-Sylow P is one of the five groups of order 8 ($C_2 \times C_2 \times C_2$, $C_2 \times C_4$, C_8 , D_4 or Q_4) and the 7-Sylow Q is one of the two groups of order 49 ($C_7 \times C_7$ or C_{49}). Hence there are 10 possibilities for G . [4]

6. Let $u = 1 + 3i + 5j + 7k$, $v = 2 + 4i + 6j + 8k \in \mathbb{H}$. We now find uv ; one way is e.g. using a table as follows:

	2	+ 4i	+ 6j	+ 8k
1	2	+ 4i	+ 6j	+ 8k
+3i	+ 6i	-12	+ 18k	- 24j
+5j	+ 10j	- 20k	- 30	+ 40i
+7k	+ 14k	+ 28j	- 42 i	- 5j

Then $uv = -96 + 8i + 20j + 20k$ so Proposition 3.1 implies a possible solution is $p = 96$, $q = 8$, $r = 20$, $s = 20$. [4]